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Robotics Formula Drill Book

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Companion to the Robotics Core Guide: one drill set per formula marked memorize there, numbered the same way.

How to drill

The goal is recall under mild pressure, not just correct answers. Cover the formula, write it from memory, then solve. Miss the recall and the set is not done, however many answers you got right.

1. **Recall first.** Before each set, write the formula and say what question it answers in one sentence.
2. **Solve by hand**, calculator allowed, no lookup. Three significant figures is enough.
3. **Check** against the answer key at the end. Mark misses.
4. **Spaced repetition**: redo every missed set the next day, then after a week. A formula is memorized when you have done its set correctly, cold, on three separate days.
5. Every set ends with one or two **design questions** that reverse the formula (given the answer, find the input). Those are the ones that build intuition; do not skip them.

Constants and conventions used throughout:

Symbol	Value
g	9.81 m/s²
1 kg·cm	0.0981 N·m
1 RPM	0.1047 rad/s
1 inch	25.4 mm
Angles	Radians unless a degree sign is written
Vectors	(x, y) or (x, y, z), right-handed, z up

Sets are numbered to match the 30-formula table at the end of the guide.

Vectors

Set 1: Dot product

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z = |\mathbf{a}||\mathbf{b}| \cos\theta$$

1. a = (3, 4), b = (1, 0). Find a·b, |a|, and the angle between them.
2. a = (1, 2, 3), b = (4, −5, 6). Find a·b.
3. Are (2, −1, 3) and (1, 5, 1) perpendicular? Show why.
4. A force F = (10, 5) N acts while an object moves d = (4, 0) m. How much work is done?
5. Project v = (6, 8) onto the direction 30° above the x axis.
6. Angle between (1, 1, 0) and (1, 0, 0)?
7. **Design.** A 3 kg robot sits on a 15° ramp. Write its weight as a vector and the up-ramp unit vector, then dot them to get the force component pulling it down the ramp.

Set 2: Cross product

$$\mathbf{a} \times \mathbf{b} = (a_y b_z - a_z b_y,\ a_z b_x - a_x b_z,\ a_x b_y - a_y b_x)$$

1. (1, 0, 0) × (0, 1, 0) = ?
2. (2, 3, 4) × (5, 6, 7) = ?
3. A 1 kg mass hangs at r = (0.3, 0, 0) m from a shoulder joint; gravity gives F = (0, −9.81, 0) N. Find the torque vector τ = r × F.
4. |a| = 2, |b| = 5, angle 30°. Find |a × b|.
5. A wheel spins at ω = (0, 0, 10) rad/s. Find the velocity of a rim point at r = (0.05, 0, 0) m.
6. Area of the parallelogram spanned by (3, 0, 0) and (1, 4, 0)?
7. **Design.** A gripper pushes down with F = (0, 0, −20) N at r = (0.2, 0.1, 0) m from the wrist. Find the torque vector and say which axis each component loads.

Trigonometry

Set 3: Law of cosines

$$c^2 = a^2 + b^2 - 2ab\cos C$$

1. a = 3, b = 4, C = 90°. Find c (and notice which theorem you just rederived).
2. a = 5, b = 7, C = 60°. Find c.
3. Links L_1 = 0.20 m and L_2 = 0.15 m meet at an elbow with interior angle 120°. How far is the fingertip from the shoulder?
4. A triangle has sides 6, 8 and 12. Find its largest angle.
5. L_1 = L_2 = 0.10 m and the target is 0.15 m from the shoulder. Find the interior elbow angle.
6. Can links of 0.10 m and 0.25 m reach a point 0.40 m away? A point 0.12 m away? Give the reachable band.
7. **Design.** L_1 = 0.25 m. Choose L_2 so that reaching 0.35 m puts the interior elbow angle at 100° (a comfortable mid-range posture).

Set 4: atan2

$$\theta = \text{atan2}(y, x) \in (-\pi, \pi]$$

1. atan2(1, 1) in degrees and radians.
2. atan2(1, −1).
3. atan2(−1, −1).
4. atan2(−2, 3).

5. A robot at (2, 3) must face the point (5, 7). What heading?
6. Explain, with the vector $(-1, -1)$, why atan(y/x) gives the wrong answer and atan2 the right one.
7. The robot's heading is 170° and the target bearing is −170°. Compute the raw difference, wrap it into (−180°, 180°], and say which way to turn.
8. **Design.** A differential-drive robot at the origin faces +y (heading 90°). The next waypoint is (−3, 3). How many degrees, and which direction, must it rotate before driving straight?

Rotations and transforms

Set 5: 2D rotation matrix

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \qquad R^{-1} = R^T$$

1. Rotate (1, 0) by 90°.
2. Rotate (1, 0) by 45°.
3. Rotate (2, 3) by 30°.
4. Write out R(60°) numerically and verify its determinant is 1.
5. Apply $R(30°)^T$ to your answer from question 3 and confirm you recover (2, 3).
6. Show that rotating by 20° then by 25° equals rotating by 45°, using the point (1, 0).
7. A lidar on a robot sees an obstacle at (2, 0) in the robot frame. The robot is at world position (1, 1) with heading 90°. Where is the obstacle in the world?
8. **Design.** An IMU is mounted rotated 180° about z relative to the body. It reports acceleration (0.5, −0.2). What does the body feel?

Set 6: Homogeneous transforms and chains

$$T = \begin{bmatrix} R & \mathbf{p} \\ 0 & 1 \end{bmatrix}, \quad T^{-1} = \begin{bmatrix} R^T & -R^T\mathbf{p} \\ 0 & 1 \end{bmatrix}, \quad {}^0T_n = T_1 T_2 \cdots T_n$$

1. T has R = I and p = (1, 2, 0). Transform the point (3, 4, 0).
2. T has R = R_z(90°) and p = 0. Transform (1, 0, 0).
3. T has R = R_z(90°) and p = (1, 0, 0). Transform (1, 0, 0).
4. Now apply the two operations of question 3 in the other order (translate by p first, then rotate). Transform (1, 0, 0) and compare.
5. T_{01} = translation (0.2, 0, 0). T_{12} = R_z(90°) with p = (0.1, 0, 0). Find frame 2's origin in frame 0, then the point (0.05, 0, 0) of frame 2 in frame 0.
6. Write the inverse of T = [R_z(90°), p = (1, 0, 0)] and verify it maps your answer from question 3 back to (1, 0, 0).
7. A camera sits 0.05 m along the tool z axis (T_tool→cam = translation (0, 0, 0.05)). The tool is at T_base→tool = [R_z(90°), p = (0.4, 0, 0.3)]. The camera sees a cup at (0.1, 0, 0.2). Where is the cup in the base frame?
8. **Design.** A robot is at world (2, 1) with heading 90°. An object is at world (2, 3). Build T_world→robot, invert it, and give the object's position in the robot frame. Sanity-check the answer with a sketch.

Calculus

Set 7: Derivatives and numeric integration

$$\frac{d}{dt}t^n = nt^{n-1}, \quad \frac{d}{dt}e^{at} = ae^{at}, \quad \frac{d}{dt}\sin\omega t = \omega\cos\omega t, \quad \dot{x}_k \approx \frac{x_k - x_{k-1}}{\Delta t}, \quad X_k = X_{k-1} + x_k\Delta t$$

1. x(t) = 3t² + 2t. Find v(t) and v(2).
2. θ(t) = 0.5 sin(4t). Find ω(t) and the maximum angular speed.
3. x(t) = 5e^{−2t}. Find v(0).
4. Differentiate t² sin t.
5. A link's horizontal position is L cos θ with θ = 2t. Find its horizontal velocity (chain rule).
6. Encoder counts sampled at 1 kHz read 100, 103, 107. Give the two velocity estimates in counts/s.
7. A robot drives at a constant 0.5 m/s for 4 s. Integrate numerically with Δt = 0.5 s and confirm the distance.
8. An accelerometer has a bias of 0.02 m/s². If you integrate it twice for 60 s, how far off is the position?
9. **Design.** An encoder read at 500 Hz jitters by 1 count. It has 4000 counts/rev on a 30 mm-radius wheel. How large is the resulting velocity spike in m/s, and what two remedies does the guide suggest?

Set 8: First-order response and time constant

$$y(t) = u\,(1 - e^{-t/\tau}), \qquad \tau_{RC} = RC, \quad \tau_{RL} = L/R, \quad f_c = \frac{1}{2\pi\tau}$$

1. R = 10 kΩ, C = 100 nF. Find τ.
2. L = 1 mH, R = 2 Ω. Find τ.
3. What fraction of the final value is reached at t = τ, 2τ, 3τ and 5τ?
4. A motor's mechanical time constant is 50 ms. How long to reach 90% of final speed?
5. A step from 0 to 10 with τ = 2 s. Find y at t = 1 s.
6. You measure a filter reaching 63% of a step after 20 ms. Find τ and the cutoff frequency.
7. **Design.** Build an RC low-pass with 5 Hz cutoff using R = 10 kΩ. Choose C.
8. **Design.** A temperature sensor has τ = 8 s. How long after power-up before the reading is within 2% of the truth?

Set 9: Natural frequency and damping

$$m\ddot{x} + c\dot{x} + kx = F, \qquad \omega_n = \sqrt{k/m}, \qquad \zeta = \frac{c}{2\sqrt{km}}, \qquad \text{overshoot} = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

1. k = 100 N/m, m = 1 kg. Find ω_n in rad/s and Hz.
2. Add damping c = 4 N·s/m. Find ζ.
3. What c gives ζ = 0.7 for the same k and m?
4. m = 2 kg, k = 800 N/m, c = 40 N·s/m. Find ω_n and ζ.
5. A P-controller on a 0.5 kg mass acts like a spring with k = K_p. Choose K_p for ω_n = 20 rad/s, then choose K_d (acting as c) for ζ = 0.7.
6. Percent overshoot for ζ = 0.5?
7. ζ = 0.2, ω_n = 10 rad/s. Find the damped frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$ and the oscillation period.
8. **Design.** You double K_p on a working joint. By what factor do ω_n and ζ change, and what will you see?

Kinematics

Set 10: Constant-acceleration motion

$$v = v_0 + at, \qquad x = x_0 + v_0 t + \tfrac{1}{2}at^2, \qquad v^2 = v_0^2 + 2a\,\Delta x$$

1. From rest, a = 2 m/s² for 3 s. Find v and distance.

2. A robot at 1.2 m/s brakes at 3 m/s². Stopping distance and time?
3. A joint accelerates from rest at 5 rad/s² for 0.4 s. Find ω and θ.
4. A robot falls off a 0.5 m table. Impact speed and fall time?
5. A drivetrain must reach 2 m/s within 1.5 m. Required acceleration?
6. A quadcopter at hover accelerates upward at 4 m/s² for 2 s. Find its climb speed and height gained.
7. A trapezoidal move: accelerate at 1 m/s² to 0.5 m/s, cruise, then decelerate at 1 m/s², covering 3 m total. Find the accel distance, cruise distance and total time.
8. **Design.** A servo joint must sweep 90° in 0.5 s using a triangular velocity profile (accelerate for half the time, decelerate for half). Find the peak angular acceleration and peak angular velocity in RPM.

Set 11: Rotation to linear

$$v = \omega r, \qquad a = \alpha r, \qquad \omega\,[\text{rad/s}] = \frac{2\pi \cdot \text{RPM}}{60}$$

1. Convert 200 RPM to rad/s.
2. A 65 mm-diameter wheel turns at 200 RPM. Ground speed?
3. You want 1 m/s on 80 mm-diameter wheels. Required wheel RPM?
4. Motor 6000 RPM through a 30:1 gearbox onto 100 mm wheels. Ground speed?
5. A 20-tooth motor pulley drives a 60-tooth wheel pulley; the motor runs 300 RPM on a 100 mm wheel. Ground speed?
6. 90 mm wheels, encoder 1440 counts/rev. The robot logged 3600 counts. Distance travelled?
7. A robot wants 1.5 m/s on 50 mm wheels. Wheel RPM? Comment on whether a typical hobby gearmotor (100–300 RPM) can do it.
8. **Design.** A 150 RPM gearmotor should give 0.6 m/s. Choose the wheel diameter.

Force, torque and inertia

Set 12: Newton's second law

$$\sum \mathbf{F} = m\mathbf{a}, \qquad \sum \tau = I\alpha$$

1. A 3 kg robot must accelerate at 1.5 m/s² on flat ground (ignore friction). Net force?
2. Same robot, same acceleration, up a 15° slope. Force required along the slope?
3. A disc with I = 0.002 kg·m² must reach α = 50 rad/s². Torque?
4. A wheel has I = 0.0005 kg·m² and 0.02 N·m of bearing friction. The motor supplies 0.1 N·m. Find α.
5. A 1.2 kg quadcopter must accelerate upward at 3 m/s². Total thrust, and thrust per motor?
6. An arm link has I = 0.05 kg·m² about its joint; gravity loads it with 1.2 N·m; the motor gives 2 N·m. Find α.
7. A 2 kg block is pulled with 10 N; μ_k = 0.3. Find its acceleration.
8. **Design.** A 5 kg robot carries 60% of its weight on the driven wheels, μ = 0.7, wheel radius 40 mm. Find the maximum acceleration before wheel slip and the total wheel torque that produces it.

Set 13: Torque from a lever

$$\tau = rF\sin\theta, \qquad 1\ \text{kg·cm} = 0.0981\ \text{N·m}$$

1. 20 N perpendicular at 0.15 m. Torque?
2. 20 N at 0.15 m, but at 60° to the lever. Torque?

3. A 2 kg payload at the end of a 0.4 m horizontal arm. Shoulder torque?
4. The same arm raised 30° above horizontal. Shoulder torque?
5. A servo is rated 25 kg·cm. Convert to N·m, then find the largest mass it can hold at 0.2 m horizontally.
6. A 0.25 m wrench with 50 N at the end. Torque?
7. A motor gives 0.3 N·m to a 40 mm-radius wheel. Push force at the ground?
8. **Design.** A link weighs 0.3 kg (center of mass at 0.15 m) and carries 0.5 kg at 0.3 m, horizontal. Find the joint torque, then the servo rating in kg·cm with a 2× safety factor.

Set 14: Moment of inertia and the parallel-axis theorem

$$I_{point} = mr^2, \quad I_{disc} = \tfrac{1}{2}mr^2, \quad I_{rod,center} = \tfrac{1}{12}mL^2, \quad I_{rod,end} = \tfrac{1}{3}mL^2, \quad I = I_c + md^2$$

1. A 0.5 kg point mass at 0.2 m. I?
2. A 0.2 kg disc of radius 50 mm about its center. I?
3. A 0.3 kg, 0.4 m rod: I about its center, then about its end (compute the end value two ways).
4. The disc of question 2 spun about a point on its rim. I?
5. An arm: the rod of question 3 pivoted at one end, plus a 0.5 kg payload at the far end. Total I about the pivot?
6. A 50 g wheel of radius 32.5 mm modeled as a disc. I?
7. A motor rotor has I = 5 × 10^{-6} kg·m² and drives through a 50:1 gearbox. Reflected inertia at the output (multiply by N²)? Compare with the arm of question 5.
8. **Design.** Two layouts for the arm of question 5: the elbow motor (0.4 kg) mounted at the shoulder axis, or at the elbow 0.4 m out. How much inertia does the second layout add, as a percentage?

Power, energy, efficiency, friction

Set 15: Power and energy

$$P = Fv = \tau\omega, \qquad E = Pt, \qquad KE = \tfrac{1}{2}mv^2 = \tfrac{1}{2}I\omega^2, \qquad PE = mgh$$

1. 0.3 N·m at 20 rad/s. Power?
2. 8 N of push at 0.5 m/s. Power?
3. Lift 2 kg by 0.5 m in 1 s. Average power?
4. Kinetic energy of a 3 kg robot at 1.2 m/s, and of a flywheel with I = 0.002 kg·m² at 100 rad/s.
5. 12 W for 2 hours, in Wh and in joules.
6. A 5 W motor runs for 30 minutes. Energy in Wh?
7. A motor delivers 0.5 N·m at 3000 RPM. Power?
8. **Design.** A 4 kg robot climbs 10 m of vertical height in 20 s. Average mechanical power, electrical power at 50% total efficiency, and energy used in Wh?

Set 16: Efficiency chains

$$\eta = \frac{P_{out}}{P_{in}}, \qquad \eta_{total} = \eta_1\eta_2\eta_3\cdots$$

1. P_in = 10 W, P_out = 7 W. Efficiency?
2. Driver 90%, motor 75%, gearbox 80%. Total?
3. The wheels need 20 W. With the chain from question 2, what does the battery supply?
4. A motor at 80% draws 2 A at 12 V. Mechanical output and heat?

5. A linear regulator drops 12 V to 5 V at 0.5 A: efficiency and heat. Repeat for a 90%-efficient buck converter.
6. Two gearboxes deliver 5 W: a worm at 50% and a planetary at 90%. Input power for each?
7. A 20 Wh battery feeds a 54%-efficient drivetrain. Usable mechanical energy?
8. **Design.** A robot needs 4 W at the wheels for 1 hour from a 3S 2200 mAh pack (24.4 Wh). Compare a planetary drivetrain (0.85 × 0.70 × 0.80) with a worm drivetrain (0.85 × 0.70 × 0.45): what fraction of the battery does each consume?

Set 17: Friction

$$F_f \le \mu_s N \text{ (static)}, \qquad F_f = \mu_k N \text{ (kinetic)}$$

1. A 3 kg robot on flat ground, μ_s = 0.6. Maximum static friction?
2. The same robot parked on a 15° ramp. Does it stay put unpowered? At what angle would it start sliding?
3. The driven wheels carry 3 kg on tile with μ = 0.45. Maximum tractive force?
4. Rolling resistance of a 5 kg robot with C_rr = 0.02?
5. Force to push a 2 kg block at constant speed with μ_k = 0.3?
6. An arm slides a 0.5 kg part across a table, μ_k = 0.4, at 0.2 m/s. Force and power?
7. You double the tire contact area. How does the friction force change, and why?
8. **Design.** A 4 kg robot wants 3 m/s² on carpet (μ = 0.5). Check for slip with four-wheel drive, then with two-wheel drive carrying half the weight; give the two-wheel maximum acceleration.

Electrical basics

Set 18: Ohm's law and power

$$V = IR, \qquad P = VI = I^2R = V^2/R$$

1. 12 V across 4 Ω. Current and power?
2. A motor winding is 0.5 Ω. Stall current and stall power at 12 V?
3. 2 A flows through a 0.1 Ω PCB trace. Voltage drop and heat?
4. Drive a 2.1 V LED at 20 mA from 5 V. Series resistor (nearest standard value) and its power?
5. A MOSFET with R_DS(on) = 20 mΩ carries 10 A. Dissipation? And at 5 A?
6. A 10 W heater on 12 V. Resistance and current?
7. 1 m of 22 AWG wire is about 0.053 Ω. Drop and loss at 5 A?
8. **Design.** A battery pack has 50 mΩ internal resistance. Your motor stalls at 24 A. How far does the pack voltage sag from 12.6 V, and why does that reset the microcontroller if it shares the rail?

Set 19: Series, parallel and the voltage divider

$$R_s = R_1 + R_2, \qquad \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}, \qquad V_{out} = V_{in}\frac{R_2}{R_1 + R_2}$$

1. 10 kΩ and 20 kΩ in series, then in parallel.
2. Three 1 kΩ resistors in parallel.
3. R_1 = 30 kΩ, R_2 = 10 kΩ across 12 V. V_out?
4. Design a divider to read a full 4S pack (16.8 V) on a 3.3 V ADC with R_2 = 10 kΩ. Choose R_1 (nearest standard value) and give V_out at 16.8 V.
5. A 10 kΩ pot across 5 V with the wiper at 30%. Wiper voltage?
6. Current and power drawn by the divider in question 3.

7. R_1 = R_2 = 10 kΩ from 10 V gives 5 V. Now connect a 10 kΩ load to the output. New V_out? What does this teach about divider loading?
8. **Design.** Divider for a 6S pack (25.2 V max) on a 3.3 V, 12-bit ADC with R_2 = 4.7 kΩ. Choose R_1 and compute the voltage resolution per ADC count.

Set 20: Batteries

$$t\,[\text{h}] = \frac{\text{Ah}}{\text{A}}, \qquad E\,[\text{Wh}] = V_{nominal} \times \text{Ah}, \qquad I_{max} = C\text{-rating} \times \text{Ah}$$

1. 2200 mAh pack drawing 1.5 A. Run time?
2. Energy in a 3S 2200 mAh LiPo?
3. A 4S 5000 mAh pack: nominal voltage, energy, full voltage and safe cutoff (3.2 V/cell)?
4. Maximum current from a 30C 2200 mAh pack?
5. A robot averages 0.8 A for logic and 2 A for motors from a 5000 mAh pack; use 80% of capacity. Run time?
6. You need 2 h at 3 A average. Minimum capacity, using 80%?
7. Eight AA NiMH cells (1.2 V, 2000 mAh) in series. Voltage and energy?
8. **Design.** A 1.2 kg quadcopter needs about 150 W to hover. Estimate flight time (80% usable) on a 3S 2200 mAh pack, then on a 4S 3000 mAh pack, and name the trade-off you cannot see in these numbers.

Signals

Set 21: ADC

$$\text{code} = \left\lfloor \frac{V_{in}}{V_{ref}}(2^N - 1) \right\rfloor, \qquad \text{resolution} = \frac{V_{ref}}{2^N}, \qquad f_{sample} > 2f_{signal}$$

1. 10-bit ADC, 5 V reference, 2.5 V input. Code?
2. 12-bit, 3.3 V reference, 1.0 V input. Code?
3. Resolution in mV for 10-bit/5 V, 12-bit/3.3 V and 16-bit/3.3 V.
4. Code 2048 on a 12-bit, 3.3 V ADC. Input voltage?
5. Through the 43 kΩ / 10 kΩ divider of Set 19, a 12-bit 3.3 V ADC reads 3000. Pack voltage?
6. A motor vibrates at 1.2 kHz. Minimum sample rate to see it? If you sample at 1 kHz, what false frequency appears?
7. A 300° potentiometer on a 10-bit ADC. Angular resolution?
8. **Design.** A strain sensor outputs 0–20 mV. How many counts does it span on a 12-bit 3.3 V ADC, and what amplifier gain would use the full range?

Set 22: PWM

$$\text{duty} = \frac{t_{on}}{T}, \qquad V_{avg} = \text{duty} \times V_{supply}$$

1. 40% duty at 12 V. Average voltage?
2. An 8-bit PWM register holds 191. Duty and average voltage at 12 V?
3. You want 7.2 V average from 12 V with 8-bit PWM. Register value?
4. A 1500 µs pulse at 50 Hz. Duty cycle?
5. A servo maps 1000–2000 µs to 0–180°. Angle for 1250 µs?
6. A timer clocked at 80 MHz generates 20 kHz PWM. Counts per period, and roughly how many bits of duty resolution?
7. Why is 490 Hz PWM a problem for a brushed motor and 20 kHz not?
8. **Design.** A 10-bit PWM on a 7.4 V rail should average 3.0 V. Register value?

Set 23: Exponential moving average

$$y_k = \alpha x_k + (1-\alpha) y_{k-1}, \qquad \alpha = \frac{\Delta t}{\tau + \Delta t}, \qquad f_c = \frac{1}{2\pi\tau}$$

1. α = 0.2, y_0 = 0, input constant 10. Compute y_1 through y_5.
2. Δt = 1 ms, τ = 10 ms. Find α.
3. Cutoff frequency for τ = 10 ms?
4. You want a 5 Hz cutoff at a 200 Hz loop. Find τ and α.
5. With α = 0.2, how many steps until the output reaches 95% of a step?
6. The filter's lag is roughly τ. For α = 0.1 at 1 kHz, what is the lag?
7. Noise of ±50 counts passes through an EMA with α = 0.1. The noise standard deviation scales by √(α/(2−α)); what is the output noise?
8. **Design.** Your PID derivative runs at 500 Hz and should ignore noise above 30 Hz. Choose τ and α.

Set 24: Complementary filter

$$\theta_k = \alpha(\theta_{k-1} + \omega\Delta t) + (1-\alpha)\,\theta_{accel}, \qquad \theta_{accel} = \text{atan2}(a_y, a_z), \qquad \tau = \frac{\alpha\Delta t}{1-\alpha}$$

1. Previous angle 20°, gyro 10°/s, Δt = 0.01 s, accelerometer angle 21°, α = 0.98. New angle?
2. Accelerometer reads a_y = 0.5 g, a_z = 0.866 g. Tilt?
3. A gyro has 0.5°/s bias. Error after 60 s of pure integration? With the filter above (α = 0.98, Δt = 0.01), the steady-state offset is αδΔt/(1−α); compute it.
4. α = 0.98 at 100 Hz. What time constant does the accelerometer correction have?
5. You want τ = 2 s at 200 Hz. Choose α.
6. The robot accelerates sideways at 0.3 g. What false tilt does the accelerometer alone report, and which term of the filter protects you?
7. Write the same filter for heading using gyro z and a magnetometer heading.
8. **Design.** A balancing robot runs at 500 Hz. Choose α for a 1 s correction time constant, and say what happens to the estimate if you accidentally use α = 0.5.

Motors and gearing

Set 25: DC motor model

$$V = IR + K_e\omega, \quad \tau = K_t I, \quad \tau(\omega) = \tau_{stall}\left(1 - \frac{\omega}{\omega_{nl}}\right), \quad P_{max} = \frac{\tau_{stall}\,\omega_{nl}}{4}$$

A motor has R = 1 Ω, K_e = K_t = 0.02 (SI units), supply 12 V. Use it for questions 1–4.

1. Stall current, stall torque and no-load speed (rad/s and RPM).
2. At 300 rad/s: back-EMF, current and torque.
3. At 25% of stall torque: speed and mechanical power.
4. Maximum mechanical power, the electrical input at that point, and the efficiency there.
5. A datasheet says stall 2 N·m, no-load 300 RPM. Torque available at 200 RPM?
6. A motor with K_t = 0.05 N·m/A draws 3 A. Torque?
7. A motor is rated 1.5 A continuous with R = 2 Ω. Copper heat at rated current, and at 12 V stall? Ratio?
8. **Design.** A wheel needs 0.13 N·m at 150 RPM. Choose the minimum stall torque and no-load speed if you want to run at 30% of stall.

Set 26: Gear ratio

$$\tau_{out} = N\tau_{in}\eta, \qquad \omega_{out} = \frac{\omega_{in}}{N}, \qquad N = \frac{T_{driven}}{T_{driving}}, \qquad I_{reflected} = N^2 I_{rotor}$$

1. 10:1 at 90% on a motor giving 0.02 N·m at 6000 RPM. Output torque and speed?
2. You need 1.5 N·m from a 0.05 N·m motor through an 85% gearbox. Minimum ratio?
3. A 12-tooth gear drives a 48-tooth gear: ratio? Two stages of 4:1 and 5:1: total?
4. 20:1 on a 5000 RPM motor. Output RPM?
5. A 30:1 worm at 50% with 0.1 N·m in. Output torque?
6. Rotor inertia 1×10^{-5} kg·m² through 20:1. Reflected inertia at the output?
7. GT2 belt (2 mm pitch), 16-tooth motor pulley, 48-tooth driven pulley. Ratio, and belt speed at 1 motor rev/s?
8. **Design.** Motor: 3000 RPM no-load, 0.1 N·m stall. Joint needs 2 N·m at 30 RPM. Running the motor at 40% of stall through an 85% gearbox, find the ratio needed for torque and the maximum ratio allowed for speed. Do they agree?

Stress and deflection

Set 27: Stress, strain, bending

$$\sigma = \frac{F}{A} = E\varepsilon, \qquad \delta_{tip} = \frac{FL^3}{3EI}, \qquad \sigma_{max} = \frac{Mc}{I}, \qquad I_{rect} = \frac{bh^3}{12}$$

Use E = 3.5 GPa for PLA (yield 50 MPa) and 69 GPa for aluminum 6061 (yield 276 MPa).

1. 500 N in tension on a 10 mm × 2 mm section. Stress in MPa?
2. A 100 mm aluminum rod stretches 0.2 mm. Strain and stress?
3. Second moment of area for a 10 mm wide × 3 mm thick bar (bending about the thin axis), in mm⁴ and m⁴.
4. A PLA cantilever of that section, 100 mm long, holds 5 N at the tip. Tip deflection?
5. Same bar in aluminum. Deflection?
6. Back to PLA, but 6 mm thick. Deflection? (Notice the factor.)
7. Maximum bending stress in the PLA bar of question 4 (M = FL at the root, c = h/2) and the safety factor against yield.
8. **Design.** Choose the PLA thickness (10 mm wide, 100 mm long, 5 N tip load) that limits deflection to 1 mm, then check the stress and safety factor at that thickness.

Control

Set 28: PID

$$u_k = K_p e_k + K_i\,\Delta t \sum_{j \le k} e_j + K_d \frac{e_k - e_{k-1}}{\Delta t}$$

1. K_p = 2, error 3. Output?
2. K_p = 2, K_i = 0.5, K_d = 0.1, Δt = 0.01 s. This tick e = 3.0, last tick e = 3.2, and this is the first tick the integral has run. Compute the P, I and D contributions and u.
3. A motor needs u = 4 just to hold position against gravity. With P-only control and K_p = 2, what steady-state error remains? Which term fixes it?
4. The actuator saturates at u = 255. The error sits at 10 for 2 s with K_i = 5. How large has the integral term grown, and what happens when the error finally reverses? Name the fix.
5. Ziegler–Nichols: the loop oscillates steadily at K_u = 8 with period T_u = 0.5 s. Starting gains (K_p = 0.6K_u, K_i = 1.2K_u/T_u, K_d = 0.075K_uT_u)?

6. A wheel's speed loop has about 5 Hz bandwidth. Minimum loop rate, and a comfortable one?
7. The setpoint jumps from 0 to 100 in one tick with K_d = 0.1 and Δt = 0.01. Derivative-of-error output on that tick? How does derivative-on-measurement avoid it?
8. **Design.** A joint carries 0.5 kg at 0.3 m. Write the gravity feedforward torque τ_ff(θ) and evaluate it at θ = 0° (horizontal) and 60°. Why does adding it let you lower K_i?

Arms and legs

Set 29: Two-link FK, IK and the Jacobian

$$x = L_1 c_1 + L_2 c_{12}, \quad y = L_1 s_1 + L_2 s_{12}, \quad \cos\theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2 L_1 L_2}$$

$$\theta_1 = \text{atan2}(y, x) - \text{atan2}(L_2 s_2,\ L_1 + L_2 c_2), \qquad J = \begin{bmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{bmatrix}, \qquad \dot{\mathbf{x}} = J\dot{\theta}, \quad \tau = J^T \mathbf{F}$$

Use L_1 = 0.20 m, L_2 = 0.15 m; c_{12} means cos(θ₁ + θ₂).

1. FK at θ₁ = 0, θ₂ = 0.
2. FK at θ₁ = 90°, θ₂ = 0.
3. FK at θ₁ = 30°, θ₂ = 60°.
4. IK for the target (0.25, 0.10): give both solutions (elbow-down θ₂ > 0 and elbow-up), then verify one with FK.
5. IK for the target (0.40, 0). What happens, and why?
6. Write J numerically at θ₁ = 0, θ₂ = 90°, and its determinant.
7. At that pose the joints move at θ = (1, 0) rad/s. Hand velocity?
8. At that pose the hand carries a downward load F = (0, −10) N. Joint torques τ = J^TF, and a one-sentence physical reading of the elbow result.
9. **Design.** A quadruped leg has thigh and shank of 0.15 m each. The foot target in the hip frame is 0.05 m forward and 0.28 m down. Find the knee angle θ₂ (relative) from the law of cosines and state whether the leg is near full extension.

Wheeled robots

Set 30: Differential drive and odometry

$$v = \frac{r(\omega_R + \omega_L)}{2}, \quad \omega = \frac{r(\omega_R - \omega_L)}{b}, \quad \omega_{R,L} = \frac{v \pm \omega b/2}{r}$$

$$\Delta s = \frac{\Delta s_R + \Delta s_L}{2}, \quad \Delta\theta = \frac{\Delta s_R - \Delta s_L}{b}, \quad x \mathrel{+}= \Delta s \cos\theta,\ y \mathrel{+}= \Delta s \sin\theta,\ \theta \mathrel{+}= \Delta\theta$$

Use r = 32.5 mm and wheelbase b = 0.20 m.

1. Both wheels at 20 rad/s. Body v and ω?
2. ω_R = 20, ω_L = 10 rad/s. Body v and ω?
3. ω_R = 10, ω_L = −10 rad/s. Body v and ω, and what motion is this?
4. Command v = 0.5 m/s, ω = 1 rad/s. Wheel speeds?
5. Turning radius for the motion in question 4 (R = v/ω)?

6. In one tick the right wheel moved 12 mm and the left 10 mm, with $\theta = 0$. Update Δs, $\Delta\theta$, x, y.
7. The robot turns 360° in place. How far does each wheel travel, and how many encoder counts is that at 1440 counts/rev?
8. You command a 360° turn from odometry and the robot physically turns 350°. Is your b too large or too small? Give the corrected b.
9. **Design.** Motors top out at 300 RPM. Maximum straight-line speed and maximum spin rate in place?

Answer key

Rounded to three significant figures. Small differences from rounding are fine; a wrong unit or sign is not.

Set 1. (1) 3; $|a| = 5$; 53.1°. (2) 12. (3) $2 - 5 + 3 = 0$, yes. (4) 40 J. (5) 9.20. (6) 45°. (7) $W = (0, -29.4)$ N, $u = (0.966, 0.259)$; $W \cdot u = -7.61$ N, i.e. 7.61 N down the ramp.

Set 2. (1) (0, 0, 1). (2) (−3, 6, −3). (3) (0, 0, −2.94) N·m. (4) 5. (5) (0, 0.5, 0) m/s. (6) 12. (7) (−2, 4, 0) N·m: 2 N·m about x, 4 N·m about y, nothing about z.

Set 3. (1) 5 (Pythagoras). (2) 6.24. (3) 0.304 m. (4) 117°. (5) 97.2°. (6) No (max 0.35 m); no (min 0.15 m); band 0.15–0.35 m. (7) $L_2 \approx 0.205$ m.

Set 4. (1) 45°, 0.785 rad. (2) 135°. (3) −135°. (4) −33.7°. (5) 53.1°. (6) atan(1) = 45° points the wrong way; atan2(−1, −1) = −135°. (7) Raw −340°, wrapped +20°, turn left (counter-clockwise). (8) Bearing 135°, rotate 45° left.

Set 5. (1) (0, 1). (2) (0.707, 0.707). (3) (0.232, 3.60). (4) [[0.5, −0.866],[0.866, 0.5]], det = 0.25 + 0.75 = 1. (5) (2.00, 3.00). (6) Both give (0.707, 0.707). (7) (1, 3). (8) (−0.5, 0.2).

Set 6. (1) (4, 6, 0). (2) (0, 1, 0). (3) (1, 1, 0). (4) (0, 2, 0): order matters. (5) (0.3, 0, 0); (0.3, 0.05, 0). (6) R = R_z(−90°), p = (0, 1, 0); it maps (1, 1, 0) → (1, 0, 0). (7) (0.4, 0.1, 0.55). (8) T^{-1} has R(−90°), p = (−1, 2); object at (2, 0): 2 m straight ahead.

Set 7. (1) $v = 6t + 2$; 14. (2) $2\cos(4t)$; 2 rad/s. (3) −10. (4) $2t \sin t + t^2 \cos t$. (5) $-2L \sin(2t)$. (6) 3000, 4000 counts/s. (7) 2 m. (8) 36 m. (9) 500 counts/s = 0.0236 m/s; low-pass before differentiating, or time between encoder edges.

Set 8. (1) 1 ms. (2) 0.5 ms. (3) 63.2%, 86.5%, 95.0%, 99.3%. (4) 115 ms. (5) 3.93. (6) $\tau = 20$ ms, $f_c = 7.96$ Hz. (7) 3.18 µF (use 3.3 µF). (8) About $4\tau = 32$ s.

Set 9. (1) 10 rad/s, 1.59 Hz. (2) 0.2. (3) 14 N·s/m. (4) 20 rad/s, 0.5. (5) $K_p = 200$ N/m, $K_d = 14$ N·s/m. (6) 16.3%. (7) 9.80 rad/s, 0.641 s. (8) $\omega_n \times 1.41$, $\zeta \div 1.41$: faster but more overshoot and ringing.

Set 10. (1) 6 m/s, 9 m. (2) 0.24 m, 0.4 s. (3) 2 rad/s, 0.4 rad. (4) 3.13 m/s, 0.319 s. (5) 1.33 m/s². (6) 8 m/s, 8 m. (7) 0.125 m accel, 2.75 m cruise, 6.5 s total. (8) 25.1 rad/s²; peak 6.28 rad/s = 60 RPM.

Set 11. (1) 20.9 rad/s. (2) 0.681 m/s. (3) 239 RPM. (4) 1.05 m/s. (5) 100 RPM, 0.524 m/s. (6) 0.707 m. (7) 573 RPM: too fast for typical gearmotors, so gear down or use a bigger wheel. (8) 76 mm.

Set 12. (1) 4.5 N. (2) 12.1 N. (3) 0.1 N·m. (4) 160 rad/s². (5) 15.4 N; 3.84 N each. (6) 16 rad/s². (7) 2.05 m/s². (8) 20.6 N → 4.12 m/s²; 0.824 N·m total.

Set 13. (1) 3 N·m. (2) 2.60 N·m. (3) 7.85 N·m. (4) 6.80 N·m. (5) 2.45 N·m; 1.25 kg. (6) 12.5 N·m. (7) 7.5 N. (8) 1.91 N·m; ×2 = 3.83 N·m ≈ 39 kg·cm.

Set 14. (1) 0.02 kg·m². (2) 2.5×10^{-4}. (3) 0.004; 0.016 both ways. (4) 7.5×10^{-4}. (5) 0.096. (6) 2.64×10^{-5}. (7) 0.0125 kg·m², about 13% of the arm. (8) +0.064 kg·m², about 67% more.

Set 15. (1) 6 W. (2) 4 W. (3) 9.81 W. (4) 2.16 J; 10 J. (5) 24 Wh = 86.4 kJ. (6) 2.5 Wh. (7) 157 W. (8) 19.6 W; 39 W; 0.22 Wh.

Set 16. (1) 70%. (2) 54%. (3) 37 W. (4) 19.2 W out, 4.8 W heat. (5) Linear 41.7%, 3.5 W heat; buck 2.78 W in, 0.28 W heat. (6) 10 W and 5.56 W. (7) 10.8 Wh. (8) Planetary 47.6% → 8.4 Wh (34% of pack); worm 26.8% → 14.9 Wh (61%).

Set 17. (1) 17.7 N. (2) N = 28.4 N, max friction 17.0 N > 7.61 N pull, stays; slides at atan(0.6) = 31°. (3) 13.2 N. (4) 0.98 N. (5) 5.89 N. (6) 1.96 N, 0.39 W. (7) Unchanged: friction depends on normal force, not area. (8) Needs 12 N; 4WD gives 19.6 N (ok); 2WD gives 9.8 N (slips), a_max = 2.45 m/s².

Set 18. (1) 3 A, 36 W. (2) 24 A, 288 W. (3) 0.2 V, 0.4 W. (4) 145 Ω → 150 Ω, 58 mW. (5) 2 W; 0.5 W. (6) 14.4 Ω, 0.833 A. (7) 0.265 V, 1.3 W. (8) 1.2 V sag → 11.4 V before wiring losses; a shared 5 V or 3.3 V regulator can drop out of regulation during the dip and the MCU resets.

Set 19. (1) 30 kΩ; 6.67 kΩ. (2) 333 Ω. (3) 3 V. (4) 40.9 kΩ → 43 kΩ; 3.17 V. (5) 1.5 V. (6) 0.3 mA, 3.6 mW. (7) 3.33 V: the load is in parallel with R_2, so use a high-impedance ADC or a buffer. (8) 31.2 kΩ → 33 kΩ; 3.14 V at 25.2 V; about 6.5 mV per count.

Set 20. (1) 1.47 h. (2) 24.4 Wh. (3) 14.8 V; 74 Wh; 16.8 V; 12.8 V. (4) 66 A. (5) 1.43 h. (6) 7.5 Ah. (7) 9.6 V, 19.2 Wh. (8) 7.8 min; 14.2 min; the heavier pack raises hover power, so real gains are smaller.

Set 21. (1) 511. (2) 1240. (3) 4.88 mV; 0.806 mV; 0.050 mV. (4) 1.65 V. (5) 12.8 V. (6) ≥ 2.4 kHz; aliases to 200 Hz. (7) 0.293°/count. (8) About 25 counts; gain ≈ 165 (use 150).

Set 22. (1) 4.8 V. (2) 74.9%, 8.99 V. (3) 153. (4) 7.5%. (5) 45°. (6) 4000 counts ≈ 12 bits. (7) 490 Hz is audible and lets current ripple; 20 kHz is above hearing and the inductance smooths it. (8) 415.

Set 23. (1) 2, 3.6, 4.88, 5.90, 6.72. (2) 0.0909. (3) 15.9 Hz. (4) τ = 31.8 ms, α = 0.136. (5) 14 steps. (6) 9 ms. (7) ±11.5 counts. (8) τ = 5.3 ms, α = 0.274.

Set 24. (1) 20.12°. (2) 30°. (3) 30°; 0.245°. (4) 0.49 s. (5) 0.9975. (6) 16.7°; the gyro term, weighted α ≈ 1. (7) ψ = α(ψ + ω_zΔt) + (1−α)ψ_mag, wrapping the difference to ±180° first. (8) α = 0.998; with α = 0.5 the time constant collapses to 2 ms and accelerometer noise and vibration dominate the angle.

Set 25. (1) 12 A; 0.24 N·m; 600 rad/s = 5730 RPM. (2) 6 V, 6 A, 0.12 N·m. (3) 450 rad/s, 27 W. (4) 36 W; 72 W in; 50%. (5) 0.667 N·m. (6) 0.15 N·m. (7) 4.5 W; 72 W; 16×. (8) Stall ≥ 0.43 N·m, no-load ≥ 214 RPM.

Set 26. (1) 0.18 N·m, 600 RPM. (2) 35.3 → 36:1 or more. (3) 4:1; 20:1. (4) 250 RPM. (5) 1.5 N·m. (6) 4×10^{-3} kg·m². (7) 3:1; 32 mm/s. (8) Torque needs N ≥ 58.8; speed allows N ≤ 60. They only just agree, with no margin: choose a stronger motor.

Set 27. (1) 25 MPa. (2) 0.002; 138 MPa. (3) 22.5 mm⁴ = 2.25×10^{-11} m⁴. (4) 21 mm. (5) 1.07 mm. (6) 2.65 mm (8× less). (7) 33.3 MPa, SF 1.5 (too low). (8) h ≈ 8.3 mm (print 8–10 mm); σ ≈ 4.4 MPa, SF ≈ 11.

Set 28. (1) 6. (2) P = 6, I = 0.015, D = −2, u = 4.015. (3) e = 2; the integral term. (4) 100; the output stays saturated after the error reverses and the joint lunges; clamp the integral at the actuator limit. (5) K_p = 4.8, K_i = 19.2, K_d = 0.3. (6) 50 Hz; 100–200 Hz. (7) 1000; differentiating the measurement instead of the error never sees the setpoint jump. (8) τ_ff =

1.47 cos θ N·m; 1.47 at 0°, 0.736 at 60°. The integral no longer has to build up the whole gravity torque, so a smaller K_i suffices.

Set 29. (1) (0.35, 0). (2) (0, 0.35). (3) (0.173, 0.25). (4) Elbow-down $\theta_2 = 80.4°$, $\theta_1 = -11.5°$; elbow-up $\theta_2 = -80.4°$, $\theta_1 = 55.1°$; FK of the first gives (0.250, 0.100). (5) $\cos\theta_2 = 1.63 > 1$: out of reach (max 0.35 m). (6) J = [[−0.15, −0.15],[0.2, 0]], det = 0.03. (7) (−0.15, 0.2) m/s. (8) (−2, 0) N·m; the elbow carries no torque because link 2 is vertical and the load line passes through the elbow. (9) r = 0.284 m, $\cos\theta_2 = 0.798$, $\theta_2 = \pm 37.1°$ (interior 143°); fairly extended, since max reach is 0.30 m.

Set 30. (1) 0.65 m/s, 0. (2) 0.488 m/s, 1.63 rad/s. (3) 0, 3.25 rad/s: spin in place. (4) ω_R = 18.5, ω_L = 12.3 rad/s. (5) 0.5 m. (6) Δs = 11 mm, Δθ = 0.01 rad, x += 0.011 m, y ≈ 0. (7) 0.628 m each; about 4430 counts. (8) Too small; b ≈ 0.206 m. (9) 1.02 m/s; 10.2 rad/s.